

The empirical recurrence for OEIS A232077, proved by transfer matrix

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Abstract

OEIS A232077 counts arrays of width 2 over $\{0, \dots, 1\}$ under a condition on each cell and its neighbours, and carries an empirical recurrence of order 4 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. Every neighbour named lies in the three rows $i-1, i, i+1$, so an ordered pair of consecutive rows is a state, the count is a walk count on $S = 20$ of them, and the conjectured recurrence is then settled by a finite exact computation. The entry writes the array with the width fixed and n counting COLUMNS; transposing it exchanges rows with columns, which exchanges the words “horizontal” and “vertical” and fixes both diagonals, and everything below is stated in that transposed frame, where n counts rows.

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1 The conjecture

OEIS A232077 is “Number of $(1+1) \times (n+1)$ 0..1 arrays with every element equal to some horizontal, diagonal or antidiagonal neighbor, with top left element zero.”. It has offset 1 and begins

$$3, 15, 46, 161, 601, 2208, 8053, 29415, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

$$\text{Empirical: } a(n) = 3*a(n-1) + a(n-2) + 4*a(n-3) + 4*a(n-4).$$

As of the “Last modified” line on the live entry (Mar 19 2018 06:57:31, revision 7) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The neighbours of a cell are the cells at the offsets

$$(-1, -1), (-1, 0), (-1, 1), (1, -1), (1, 0), (1, 1),$$

those falling outside the array being absent. The entry requires that every cell equals at least one of its neighbours. The entry also fixes the top left cell to zero, which restricts the states an array may begin in and nothing else.

3 The count is a walk count

Every offset above moves the row index by at most one, so whether a cell of row i passes depends on rows $i-1$, i and $i+1$ and nothing else. Take as vertices the ordered PAIRS (p, c) of consecutive rows, allowing p to be ABSENT — that is what lets a walk of no steps describe a one-row array — and put an edge $(p, c) \rightarrow (c, x)$ exactly when every cell of the middle row c passes with p above it and x below it. A state with p absent may begin an array; a state (p, c) accepts when every cell of c passes with nothing below it. An array of L rows is then a walk of $L - 1$ steps, each step settling one row, and every array arises from exactly one walk. Hence

$$a(n) = \iota^\top M^j \tau$$

with ι the indicator of the beginning states, τ of the accepting ones, and j the walk index, which differs from the entry's own n by a constant read off from its published terms. There are $S = 20$ states, and a is C -finite.

At $n = 1$ the array has one row, and the condition is tested there with nothing above or below it; the entry's first term is 3, which the model reproduces along with every later term. That is the cheapest check that the neighbour set and the boundary have been read the same way the entry reads them: a condition tested as though the missing rows were present, rather than absent, gives a different first term. There are $2^2 = 4$ possible rows over the entry's own alphabet. The state count $S = 20$ is exactly $4 \cdot 5$, one state for each ordered pair of a row and its possible predecessor, the absent predecessor included.

4 The proof

Lemma 1. *Let $q(t) = t^4 - \sum_i c_i t^{4-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+4} - \sum_i c_i M^{j+4-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) = 3a(n-1) + a(n-2) + 4a(n-3) + 4a(n-4)$$

for every $n > 4$.

Proof. The vector $w = q(M)\tau$ was formed by 4 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 4 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 24 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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